THE BONDI DIPOLE IN FULL NUMERICAL RELATIVITY: A SELF-ACCELERATING POSITIVE–NEGATIVE MASS BINARY

NIKITA M. SHIROKOV*
Independent Researcher, Moscow, Russia
(Dated: August 17, 2026)

Bondi observed in 1957 that bodies whose active gravitational masses have opposite signs should *self-accelerate*: the negative one chases the positive one it repels, and the pair runs off together. Seven decades of analyses treated only point particles or exact metrics; here we present the first 3+1 numerical-relativity evolutions of a "Bondi dipole," built on what is, to our knowledge, the first stable, asymptotically flat star with negative total (ADM) mass dynamically evolved in numerical relativity—an object that proved the most robust in the campaign, its reversed gravity diluting rather than re-accreting its own shed field. Two complex scalar fields share identical dynamics; only the phantom sector sources Einstein's equations with a minus sign: inertial and passive masses positive, active mass negative. Released at rest, the mixed pair—and only the mixed pair—self-accelerates as Bondi predicted, at 50 to 150 times the residual motion of its single-star controls and at least 20 times that of its same-sign pairs. The drift reverses when the stars swap places, scales with the phantom's mass to 0.6–3.4% across the clean window, and approaches the Newtonian rate at wider separations. In the signed bookkeeping the accelerating pair should carry near-zero total momentum, as in Forward's analysis; the direct momentum-budget measurement is deferred to follow-up work.

Draft — required before submission: (1) convergence tests: rerun `PM` and one control at $N = 192, 256$ (quadrant symmetry funds 256), Richardson-extrapolate the drifts, quote a convergence order and numerical error bars on every headline percentage; (2) run the per-sector momentum-balance diagnostic ($P_i^{\mathrm{signed}}(t)$, Sec. VII) and re-run the separation series with the core-weighted centroid; (3) sponge layer *and* larger domain ($L \geq 128$) for the reflected radiation bath, extending the window past $t = 60$ for the constant-gap test; (4) force refinement on the scalar cores from $t = 0$—the χ-curvature tagger never fires inside the window (Appendix A); tag on the field profiles, target core $\Delta x \leq 0.125$; (5) a robustness point at a second frequency or coupling; a phase-offset and an unequal-mass `PP` control; (6) regenerate figures with distinct line styles (no gray-on-gray) and remove the stray "1" panel artifacts. Full plan: `research/bondi_dipole/Debug.md`.

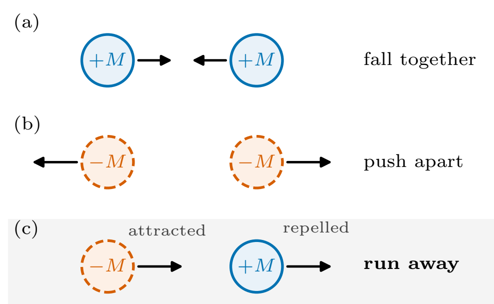

FIG. 1. Bondi's sign rules for two bodies released at rest, in the style of Forward's analysis [5]. Every body here has positive inertial and passive mass; only the *active* (source) mass carries a sign—negative for the dashed bodies. (a) Two positive-mass stars attract each other and fall together; (b) two negative-mass stars repel each other and push apart. In both cases the forces oppose, and the pair's centre stays put. (c) The mixed pair—the Bondi dipole: the positive star is repelled by the negative one, the negative star is attracted by the positive one, so both forces point the *same* way and the pair runs off together, carrying zero total momentum in the signed bookkeeping. Each row is realized as a run of the matrix in Table I.

I. INTRODUCTION

Bondi [1] gave negative mass its modern treatment, distinguishing inertial, passive, and active gravitational mass and noting the striking consequence of mixing signs: a negative-mass body is *attracted* by a positive one, the positive body is *repelled* by the negative one, and the pair runs off together (Fig. 1)—momentum conserved, because the negative body's contribution is negative. Bonnor and Swaminarayan [2] found exact solutions for uniformly accelerated opposite-mass pairs, later generalized in Ref. [3]; Bonnor [4] asked why mass is positive and found no answer in the theory. Forward [5] analyzed the Newtonian two-body problem exhaustively—for equal and unequal mass magnitudes alike—showing "in tedious detail" that the self-accelerating pair violates neither momentum nor energy conservation; objections to negative mass must be sought elsewhere [6, 7].

Why does this deserve a full numerical-relativity campaign? Because nothing in Einstein's equations forces mass to be positive—positivity is a theorem only once energy conditions are imposed on the matter: for asymptotically flat data obeying the dominant energy condition the ADM mass is provably nonnegative [24, 25], so an asymptotically flat negative-mass body *must* violate an energy condition (with de Sitter asymptotics, remarkably, it need not [26])—and modern physics keeps pro-

* shirokov.nm@phystech.edu

ducing sources that strain those conditions: Casimir energies, quantum fields in curved spacetime, a dark-energy sector whose observational fits tolerate phantom equations of state [8]. Every traversable wormhole [9] and warp spacetime [10] requires matter of exactly this kind, and the standing objection to all of them is Bondi's runaway itself: matter that accelerates for free looks like a perpetual-motion machine, hence absurd [6, 7]. That objection has only ever been examined for point particles and special exact metrics. Whether the runaway actually happens in full nonlinear general relativity—with finite bodies, backreaction, and radiation—and whether momentum conservation really survives it, is a concrete, checkable question. Answering it also requires learning to evolve energy-condition-violating matter stably in numerical relativity at all, a capability any quantitative study of exotic spacetimes will need.

All of the analyses above, seven decades of them, treat point particles, prescribed metrics, or frozen mass parameters. The nonlinear evolutions that do exist for energy-condition-violating matter—ghost-scalar wormhole spacetimes, in spherical symmetry [12, 27]—concern single objects, and the phantom-field compact stars of Ref. [28] are static constructions. None of these can say whether such a configuration can be posed as a *consistent initial-value problem*—constraint-satisfying initial data for dynamical matter—or whether the runaway survives full nonlinear evolution with backreaction, radiation, and finite body size. That is the gap this paper closes; we are not aware of any previous 3+1 evolution of a gravitationally bound positive–negative active-mass pair with dynamical, constraint-solved matter.

Our contributions:

(i) A minimal field-theoretic realization of Bondi's sign structure: a *bicomplex* scalar model in which both sectors obey identical Klein–Gordon dynamics and only the Einstein source is signed (Sec. II).

(ii) Dressed soliton stars for both signs, solved at fixed frequency as constraint-satisfying initial data—including, as far as we know, the first asymptotically flat star with negative total (ADM) mass dynamically evolved this way (Sec. III); static phantom-field stars [28] and de Sitter negative-mass bubbles [26] were known.

(iii) A falsifiable run matrix—mixed pair, same-sector pairs, single stars, a mirrored pair, two wider separations—run under identical numerics and verified at $t = 0$ against the independently solved stars (Secs. IV, V).

(iv) The result: the mixed pair, and only the mixed pair, self-accelerates—with unequal mass magnitudes (the reference pair, $|M_-|/M_+ = 1.20$) and with the magnitudes matched, echoing Forward's unequal-mass cases; the effect scales with the source mass and converges to the Newtonian point-mass prediction as the bodies separate (Sec. V).

(v) An honest accounting of failure modes and diagnostic biases that mimic or mask the physics under test (Secs. III D, VI).

One more first rode along with the physics: every evolution here ran on a single GPU, with `GRTeclyn` [21], the AMReX-based [23] GPU port of `GRChombo` [20]. Together with the ghost-scalar wormhole collapse campaign of Ref. [36], run with the same code, these are to our knowledge the first evolutions of energy-condition-violating matter in GPU-native numerical relativity, and the fact that the campaign—initial data, nine production runs, and their diagnostics—fit on one multi-GPU node is itself a statement about how accessible this class of question has become.

Throughout, $G = c = 1$ and the scalar mass sets the unit, $m = 1$. Drifts along the pair axis are quoted as $\Delta x = x(t) - x(0)$, and d denotes the instantaneous barycentre separation. The paper follows the build order: Sec. II defines the matter model, Sec. III constructs the stars and the initial data—including the failed attempts that shaped the recipe—Sec. IV lays out the runs, Sec. V the measurements, and Secs. VI–VII their limits and meaning.

II. THE BICOMPLEX SCALAR MODEL

We first need matter that can realize Bondi's sign structure while remaining an honest field theory, and the requirements are stiffer than they look. The bodies must keep canonical *dynamics*—positive inertial and passive mass—while only their gravitational *source* changes sign; they must form stable, localized stars; and the negative-mass star must be held together by something other than gravity, because its own gravity pushes it apart—the self-repulsion of negative-mass material that Forward's analysis already identified [5]. Each obvious alternative fails one of these tests. The negative-mass Schwarzschild spacetime is a naked singularity with unstable perturbations [11], and contains no matter from which to build constraint-satisfying initial data. A negative-energy fluid has no equilibrium to sit in: a fluid star is bound by the very attraction being flipped. A ghost scalar—kinetic term reversed in the Lagrangian—does source gravity negatively, but its field dynamics reverse too, so its inertial and passive masses flip along with the active one (Forward's all-negative matter [5], not the active-only signature under test), and such fields, evolved in numerical relativity as wormhole support, either collapse or explode [12, 27].

What passes is a *soliton*: matter self-bound by its own interactions, for which gravity is a dressing rather than the binding agent. We therefore take two independent complex scalar fields, Φ_+ (*canonical*) and Φ_- (*phantom*), living on one spacetime. Both are governed by the same potential, the standard sextic form of Q-ball and solitonic boson-star studies [13–15],

$$V(|\Phi|^2) = \tfrac{1}{2}m^2|\Phi|^2 - \tfrac{1}{4}\lambda|\Phi|^4 + \tfrac{1}{6}\mu|\Phi|^6, \tag{1}$$

where m is the field mass and $\lambda, \mu > 0$ are self-coupling constants. Each term has a fixed job: the mass term makes the field oscillate, the attractive quartic ($-\lambda$) lets the field clump into localized, self-bound lumps, and the repulsive sextic ($+\mu$) stops the clumping short of collapse—with the quartic alone, which is critical in three dimensions, a lump either collapses or disperses. With all three terms the flat-space theory supports Q-ball–type solitons. ("Lump" is our generic word for such a self-bound blob of field; it becomes a "star" once gravity dresses it.) Self-binding is what lets the phantom star exist at all: a gravity-bound body cannot survive having its gravity reversed, but a field-bound lump merely gets dressed—compressed by attractive gravity in one sector, dilated by repulsive gravity in the other (Sec. III). Sharing one potential is deliberate, not economy: it makes the matter physics of the two sectors *identical*, so that the single sign introduced next is the only difference between them—and therefore the only possible cause of any asymmetry the simulations show.

The entire sign structure lives in one place: the Einstein equations are sourced by the *signed* sum of two canonical stress tensors,

$$G_{\mu\nu} = 8\pi\left(T_{\mu\nu}[\Phi_+] - T_{\mu\nu}[\Phi_-]\right), \tag{2}$$

while the field equations of *both* sectors keep their canonical form,

$$\Box\,\Phi_\pm = 2\,\frac{\partial V}{\partial|\Phi|^2}\,\Phi_\pm, \tag{3}$$

the factor of 2 fixed by the $\frac{1}{2}$-normalized kinetic term of the sector Lagrangian below. In the 3+1 language the matter sources seen by normal observers—the energy density ρ and momentum density S_i that feed the ADM constraint and evolution equations—are accumulated per sector with a sign $\epsilon_\pm = \pm 1$:

$$\rho = \sum_{s=\pm} \epsilon_s\,\rho[\Phi_s], \qquad S_i = \sum_{s=\pm} \epsilon_s\,S_i[\Phi_s], \tag{4}$$

and likewise for S_{ij}, each bracket being the standard scalar-field expression.

Three consistency statements follow from the structure, and are worth making explicit. The model derives from the action

$$S = \int d^4x\,\sqrt{-g}\left[\frac{R}{16\pi} + \mathcal{L}[\Phi_+] - \mathcal{L}[\Phi_-]\right], \tag{5}$$

with the canonical $\mathcal{L}[\Phi] = -\frac{1}{2}\nabla_\mu\Phi^*\nabla^\mu\Phi - V(|\Phi|^2)$: the overall sign of a sector's Lagrangian cancels from its Euler–Lagrange equations—hence the canonical dynamics of Eq. (3)—but survives the metric variation—hence the signed source of Eq. (2). Bianchi consistency is then automatic: each sector's stress tensor is separately conserved on shell, so the signed sum sourcing $G_{\mu\nu}$ is conserved too. And well-posedness is inherited: the principal part of CCZ4 plus two canonical Klein–Gordon fields is exactly that of the standard system—the sector sign enters only algebraic source terms—so strong hyperbolicity is unaffected. What is *not* claimed is quantum viability: a sector entering gravity with negative energy admits graviton-mediated vacuum decay into positive–negative pairs [29, 30]; the model is a classical probe, in the sense of the remarks below.

This structure delivers exactly the mass signature Bondi's argument needs. How the phantom *moves* is set by its field equation (3), which is completely ordinary: the phantom field resists acceleration and falls toward gravitational wells like any other matter—in Bondi's vocabulary, its *inertial* and *passive* masses are positive. How the phantom *gravitates* is set by the minus sign in Eq. (2): the field it sources is that of a negative mass, so its *active* mass is negative and it repels everything. In particular its Arnowitt–Deser–Misner (ADM) mass—the total mass a distant observer would infer from the far gravitational field—comes out negative.

Now place one star of each kind side by side. The canonical star attracts the phantom, which falls toward it; the phantom repels the canonical star, which retreats. Both accelerations point the same way, from phantom toward canonical, and the pair runs off together. This is the precise sense in which "Newton's third law fails" here: the two forces point in the *same* direction rather than opposing each other. No conservation law breaks. The conserved momentum belongs to the total signed stress-energy of Eq. (4), in which the phantom's momentum enters with a minus sign; the two growing momenta cancel, and the accelerating pair carries near-zero total momentum—the cancellation Forward exhibited for the Newtonian problem [5], here general-relativistic.

Two honest remarks. The phantom sector violates the null energy condition by construction; this is a model of *if such matter existed, what would gravity do*, in the spirit of phantom cosmology [8], not a claim that it exists. And our sign structure differs from Forward's all-four-masses-negative matter [5]: here only the source sign flips. The two-body phenomenology is the same, because passive and inertial mass cancel—each body's acceleration is set by the *other* body's active mass, the property under test.

The model is implemented in `GRTeclyn` [20, 21, 23], evolving CCZ4 [19] plus the two complex fields, with Eq. (4) accumulated star by star in the initial-data solver `GRTresna` [22] so that one constraint solve can carry mixed-sign matter.

III. INITIAL DATA: GRAVITATIONALLY DRESSED SOLITONS

The task of this section is to turn the model into two stars on one initial slice: fix the couplings, solve each star as a spherical equilibrium, transfer the solved profiles onto the evolution grid, and solve the constraints for the pair. None of these steps is optional—the potential alone, without the gravitational dressing and the exact seeding below, produces no star that survives evolution. The failed campaigns that established this the hard way close the section.

A. Couplings

The potential (1) has three parameters: the mass $m = 1$ (our unit) and the self-couplings λ, μ. One more number needs a name before they can be chosen. A soliton of this theory oscillates coherently, $\Phi \propto e^{-i\omega t}$, and the frequency ω labels the solution family: $\omega = m$ is the free-field limit, the deficit $1-\omega$ is the binding energy per unit mass, and lower ω means a more tightly bound, more massive lump.

Two combinations of the couplings control the physics. The dimensionless ratio λ^2/μ sets the *shape* of the potential and, through it, the lowest frequency a flat-space soliton can have: the classic Q-ball existence bound $\omega^2_{\rm min} = \min_{|\Phi|}\left[2V/|\Phi|^2\right]$ [14] evaluates, for the potential (1), to $\omega_{\rm min} = \sqrt{1 - 3\lambda^2/16\mu}$, attained at the field value $\sqrt{3\lambda/4\mu}$—which is also the interior ("thin-wall") amplitude of a large soliton. All runs use $\lambda^2/\mu = 4.8$, so $\omega_{\rm min} = 0.316$ and the working frequency $\omega = 0.55$ sits comfortably inside the allowed range, at 45% binding. With the shape fixed, the remaining freedom—the overall strength of the couplings—sets that amplitude, and with it how strongly the lump gravitates: scaling λ and μ up together at fixed λ^2/μ makes the field fainter and its self-gravity weaker. The values used everywhere, $\lambda = 10240$ and $\mu \simeq 2.18 \times 10^7$ (amplitude $\simeq$0.019), were forced on us: every more strongly gravitating choice produced stars that collapsed or dispersed (Sec. III D), and these couplings are the first at which a dressed $\omega = 0.55$ star exists at all.

B. The dressed-star problem

The problem, stated plainly: a flat-space soliton solves the field equation with gravity switched off, and such an object cannot simply be placed on an initial slice. It fails as initial data, because the Hamiltonian and momentum constraints know about the field's energy and are violated by a bare Q-ball; and it fails as an equilibrium, because once its own gravity acts the profile is no longer an eigenstate—seeded anyway, the star breathes, sheds, and dies (Sec. III D documents exactly this). The star we need is solved with its gravity *on*: field and metric adjusted together until the profile is the eigenstate of the very spacetime it curves. That is what "dressed" means here.

Dressing the soliton in its own gravity is the boson-star problem [16, 17] with a sextic self-interaction (see Ref. [18] for a review), posed for both source signs. Inserting the harmonic ansatz $\Phi = \varphi_0(r)\, e^{-i\omega t}$ and a conformally flat, maximally sliced metric $ds^2 = -\alpha^2 dt^2 + \psi^4 \delta_{ij} dx^i dx^j$ into Eqs. (2)–(3) reduces the problem to three coupled ordinary differential equations in r (the standard boson-star reduction [18]),

$$\begin{aligned} \psi'' + \tfrac{2}{r}\psi' &= -2\pi\psi^5\rho, \\ \alpha'' + \left(\tfrac{2}{r} + \tfrac{2\psi'}{\psi}\right)\alpha' &= 4\pi\psi^4\alpha\,(\rho + S), \\ \varphi_0'' + \left(\tfrac{2}{r} + \tfrac{\alpha'}{\alpha} + \tfrac{2\psi'}{\psi}\right)\varphi_0' &= \psi^4\left(\frac{dV}{d\varphi_0} - \frac{\omega^2}{\alpha^2}\varphi_0\right), \end{aligned} \tag{6}$$

with $\rho = \frac{1}{2}\omega^2\varphi_0^2/\alpha^2 + \frac{1}{2}\psi^{-4}\varphi_0'^2 + V$ and $\rho + S = 2\omega^2\varphi_0^2/\alpha^2 - 2V$; here $dV/d\varphi_0 \equiv 2\varphi_0\,\partial V/\partial|\Phi|^2$ is the derivative of the effective real-field potential $V(\varphi_0^2)$, the factor of 2 matching Eq. (3). **The phantom star is obtained by flipping the sign of ρ and $\rho + S$ in the metric equations only**; the Klein–Gordon line keeps its canonical form, mirroring Eqs. (2)–(3). The result is a self-interaction-bound soliton with repulsive gravity: $\psi < 1$, $\alpha > 1$ in the core (reversed for the canonical star) and negative ADM mass.

We integrate Eqs. (6) by shooting; the solver and the $M(\omega)$ family scan behind Fig. 2 ship in the results pack. Solving at *fixed frequency* matters: heavy-branch sextic stars sit a fraction of a percent below the effective-potential top, so at fixed central amplitude the eigenvalue in ω is an exponentially thin needle, and a shooter on ω lands on the light branch (a request for the $\omega = 0.55$ star returned a 5× lighter $\omega = 0.75$ star). Instead we bisect the central amplitude φ_c at fixed integration-frame frequency and outer-iterate until the rescaled frequency ω/α_∞ matches the request, to $\sim$$10^{-5}$.

Figure 2 shows both families. The reference pair uses the two $\omega = 0.550$ stars: canonical ($\varphi_c = 0.019695$, $M_{\rm ADM} = +0.063951$, central lapse 0.9773) and phantom ($\varphi_c = 0.019589$, $M_{\rm ADM} = -0.076962$, central lapse 1.0249). Because both are taken at the *same* frequency, and the phantom branch is the heavier there, the reference pair is intrinsically unequal: $|M_-|/M_+ = 1.203$. For the equal-$|M|$ run of Sec. IV the phantom is instead solved at $\omega = 0.56598$, where $M_{\rm ADM} = -0.063950$ matches the canonical star's mass to one part in 10^5. Matching the *ADM* masses—rather than bare field amplitudes or Noether charges—is the choice Bondi's point-mass limit dictates: each body's exterior field, and hence the force it exerts on its partner, is set by its ADM mass alone. The different internal frequency this requires (0.56598 against 0.550) is part of the phantom's equilibrium at that mass, not an extra knob.

C. From radial profiles to three-dimensional initial data

The solved profiles $\varphi_0(r)$, $\alpha(r)$, $\psi(r)$ are next transferred onto the Cartesian evolution grid, one star at

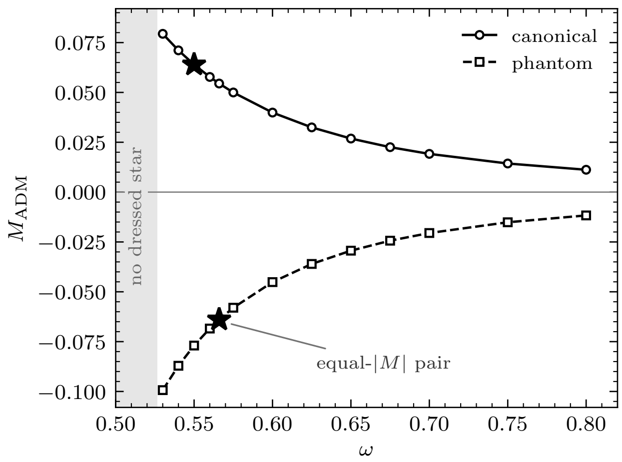

FIG. 2. ADM masses of the spherical stars against the field oscillation frequency ω, from Eqs. (6) at the working couplings: canonical stars ($M_{\mathrm{ADM}} > 0$, circles) and phantom stars ($M_{\mathrm{ADM}} < 0$, squares). Lower frequency means deeper binding and larger $|M_{\mathrm{ADM}}|$; in the shaded region the solver finds no gravitationally dressed star. At equal frequency the phantom star is the heavier in magnitude (20% at $\omega = 0.55$, 4% at $\omega = 0.80$), its repulsive self-gravity spreading it over more volume. Stars ($\star$): the equal-$|M_{\mathrm{ADM}}|$ pair of run `PM-eq`, matched to one part in 10^5.

each release position. Two details of this step mattered enormously (Sec. III D). First, the seeded amplitude must be the star's own solved φ_c—any rescaling de-tunes the eigenstate. Second, a boson star is stationary but not static: its phase rotates, so the field carries a $U(1)$ momentum, and the momentum conjugate to Φ is defined per unit *proper* time of the normal observers, $\Pi = \alpha^{-1}\partial_t\Phi$ for zero shift. The harmonic phase therefore seeds

$$\Pi_{\mathrm{im}} = -\frac{\omega}{\alpha(r)}\,\varphi_0(r); \tag{7}$$

a flat-space seed ($\alpha \equiv 1$) would start the star slightly off its eigenstate everywhere the lapse differs from one, which is why the star tables carry $\alpha(r)$. Each star is seeded from its own table with its own frequency; giving the $\omega = 0.56598$ star a global $\omega = 0.550$ phase velocity would likewise de-tune it.

`GRTresna` then solves the constraints for the pair (CTTKHybrid scheme [32], sources summed with their signs star by star) to 0.1%; at an earlier 1% tolerance the leftover residual radiated away as a ring in χ that shook the stars' envelopes. The evolution code keeps the solved metric and reconstructs the matter fields analytically from the same tables.

One superposition detail deserves stating, because for scalar lumps it is physical rather than cosmetic. All pair runs seed the two lumps with $U(1)$ phases 0 and π. For the mixed pair the choice is inert—the sectors are independent fields, so their relative phase is pure gauge—but in the same-sector controls both lumps live in *one* complex field, where relative phase changes the interaction qualitatively (anti-phase pairs of scalar solitons repel where in-phase pairs attract), and plain superposition of boson-star data is known to inject spurious artifacts that a constraint re-solve reduces but does not eliminate [31]. The full CTTKHybrid solve to 0.1% bounds this systematic here; a phase-offset `PP` rerun to measure it directly is future work.

D. How the matter was made to hold together

None of the choices above were right on the first try. Before any runaway could be claimed, a single lump had to hold its own size; four campaigns failed that gate. Heavy lumps collapsed ($\min\chi \to 0.009$ by $t \approx 25$), light lumps evaporated (rms $5 \to 30$ by $t \approx 50$), and lumps of intermediate weight did both—indicting the *shape* of the seed, not its weight. Four root causes, in order:

1. **The amplitude clamp.** The seeding script clamped the amplitude to the flat-space thin-wall value $\sqrt{3\lambda/4\mu}$, 4.5% below the solved eigenstate: an exact Q-ball shape at the wrong height is not a Q-ball, and it breathed, shed, and evaporated. *Fix:* seed at the solved φ_c.

2. **No dressed equilibrium existed.** The corrected flat-space seed still dispersed: a flat-space eigenstate is an equilibrium only in flat space, and the dressed family showed the target star did not exist at the original couplings. *Fix:* weaken the couplings to the working values and seed the dressed profile with its own lapse, Eq. (7).

3. **Two seeding paths disagreed silently.** The constraint solve used the tabulated star while the evolution's matter reconstruction fell through to a Gaussian: the solved metric backed one object, the evolution started another. *Fix:* both paths resolve the same star, enforced to 10^{-10} by a regression test.

4. **The phantom path had never run.** An old veto downgraded exotic lumps to canonical matter whenever self-gravity was on—correct for a gravity-*bound* star, irrelevant for one bound by self-interaction. *Fix:* lift the veto. The first phantom star evolved matched its predicted profile, with the predicted mirror-image metric signature (χ just *above* unity).

Two residual effects were attributed rather than fixed: the $\pm 8\%$ radial breathing is resolution-intrinsic (resolving with $30\times$ tighter momentum residuals reproduced it crest for crest), and the rms-radius stream tracks the shed radiation bath (Sec. VI), not star health.

The exportable lesson: *verify the seed before believing the evolution.* Every run was checked at $t = 0$ against the independently integrated star solution; two of the four root causes were caught by that check, not by watching movies.

IV. EVOLUTION SETUP AND THE RUN MATRIX

With both stars in hand, the experiment is to release them at rest and watch the barycentres. Evolutions use `GRTeclyn` (one GPU per run, single rank): CCZ4 [19] with damping $(\kappa_1, \kappa_2, \kappa_3) = (3, 0, 1)$, 1+log slicing [33] and a Gamma-driver shift [34, 35] with $\eta = 1$, fourth-order spatial stencils, RK4 time stepping, Kreiss–Oliger dissipation $\sigma = 2$, and $\Delta t = 0.02\,\Delta x$ (deliberately conservative, roughly a tenth of the usual Courant choice, and not tuned; cost was not a constraint at this resolution). The domain is $L = 64$ at base resolution $N = 128$ ($\Delta x = 0.5$), with Sommerfeld (radiative) outer boundaries and wave-extraction shells at $R = 8$ and 16. One mesh-refinement level is allowed (`max_level` = 1, tagging on the curvature of the conformal factor, $\Delta x\,\|\partial_i\partial_j\chi\| \geq 0.02$, regrid every 16 steps), but these stars are weak-field objects—$|\chi - 1| \lesssim 0.02$ over scales of several m^{-1}, an order of magnitude below threshold—so the criterion first triggers only at $t \approx 48$, when the deepening canonical well steepens past it: after the measurement window. Every quantitative result was therefore produced on the uniform grid (Appendix A). Stars are seeded at $x = \pm 4$ about the domain centre ($d_0 = 8$, about 1.6× their rms radius of ∼5), *at rest*—no boosts, no velocities, no driving of any kind—so any drift is gravitational.

Runs are named by content: `PM` is the mixed canonical–phantom pair, `PP` and `MM` the same-sector pairs, `P` and `M` the single stars, `MP` the mirrored pair (sectors swapped in place), and `PM-eq` the mixed pair rerun with the phantom's mass magnitude matched to the canonical star's, $|M_-| = M_+$ to one part in 10^5 (Fig. 2).

The primary diagnostic is the per-sector barycentre stream, sampled continuously: for each sector $s = \pm$,

$$\bar{x}_i^{(s)}(t) = \frac{\int_{\mathcal{D}} w_s\, x_i\, d^3x}{\int_{\mathcal{D}} w_s\, d^3x}, \qquad w_s = \left(|\Phi_s|^2 + |\Pi_s|^2\right)^{1/2}, \tag{8}$$

together with the total weight $\int w_s\, d^3x$ and the rms radius about $\bar{x}^{(s)}$. The accounting region $\mathcal{D}$ is the full computational domain in coordinate volume; its finite extent, and the coordinate (gauge-dependent) nature of the centroid, are the source of the biases quantified in Sec. VI. Constraint norms (Appendix A), confinement (peak amplitude, confined fraction, $\min\chi$), Ψ_4 modes (recorded but not used quantitatively: both extraction shells sit inside the matter distribution), and energy-condition monitors are recorded alongside.

Table I is the experiment: each row is a falsifiable prediction of Sec. II's sign structure—a same-sector pair or lone star that drifted like the mixed pair would kill the claim. `PM-eq` changes exactly one physical quantity (the phantom's mass, by 17%); `MP` reflects the geometry; the wider separations probe the force law.

TABLE I. The falsifiable run matrix. The sign structure of Sec. II predicts co-drift in $+x$ for the mixed pair, reversal under mirroring, a weaker runaway at wider separation, and *no* net drift for same-sector pairs and lone stars. Every run was solved to Ham ≤0.09%, Mom ≤0.08% (Appendix A) and verified at $t = 0$ against the independently solved star profiles (within 0.2%). Drifts in code units at $t_{\rm end}$ ($t = 40$ for singles, 60 for pairs).

run	drift Φ_+	drift Φ_-	verdict
`PM` $(+,-)$	+3.35	+9.56	**runaway**
`PM-eq` (equal $\|M\|$)	+2.94	+9.59	runaway
`MP` (mirror)	−3.23	−9.07	reversed
`PP` $(+,+)$	+0.16	—	null
`MM` $(-,-)$	—	+0.04	null
`P` (single +)	+0.06	—	null
`M` (single −)	—	+0.07	null
`PM`, d_0=12	+0.72	+2.78	weaker
`PM`, d_0=16	−0.34[a]	+0.74	weaker

[a] Wrong-signed and biased: at this separation the phantom's repulsion pushes canonical matter through the $+x$ boundary, and the lost leading edge drags the field-weighted centroid of Eq. (8) backwards (Sec. VI); the phantom column is the trustworthy one here.

V. RESULTS

A. The runaway

Figures 3 and 4 carry the headline. In `PM`—canonical star at $x = +4$, phantom at $x = -4$, both at rest—*both* barycentres move in $+x$: by $t = 60$ the canonical star has moved +3.35 and the phantom +9.56, closing the gap from 8.00 to 1.79. The four controls, run under identical numerics, do not move (+0.16 and +0.04 for the same-sector pairs, +0.06/+0.07 for the singles); taking the single-star drifts as the noise floor, the mixed pair exceeds it by 50 to 150 times, and even the largest control residual—the slowly merging `PP` pair's +0.16—by a factor of 20 or more, with constraint norms at the level of the non-drifting controls throughout the window (Appendix A).

The barycentre *speeds* (Fig. 5) say the same thing kinematically: both bodies accelerate together from rest—a sustained pull, not an impulse. (Speeds, like everything else, are in code units with $c = 1$, so $v_x = 0.44$ means $0.44c$.) The canonical star reaches $v_x \approx 0.06$ at the window edge $t = 30$ and peaks near 0.12; the phantom passes 0.11 at $t = 30$ and reaches $v_x \approx 0.44$, a genuinely relativistic infall, as the pair nears contact. (The late decline of the canonical curve is the halo bias of Sec. VI, not a deceleration.)

Why does the phantom cover almost three times the distance? Not because of the mass imbalance—the run matrix contains the direct test. In `PM-eq` the mass magnitudes are matched to one part in 10^5, and the phantom

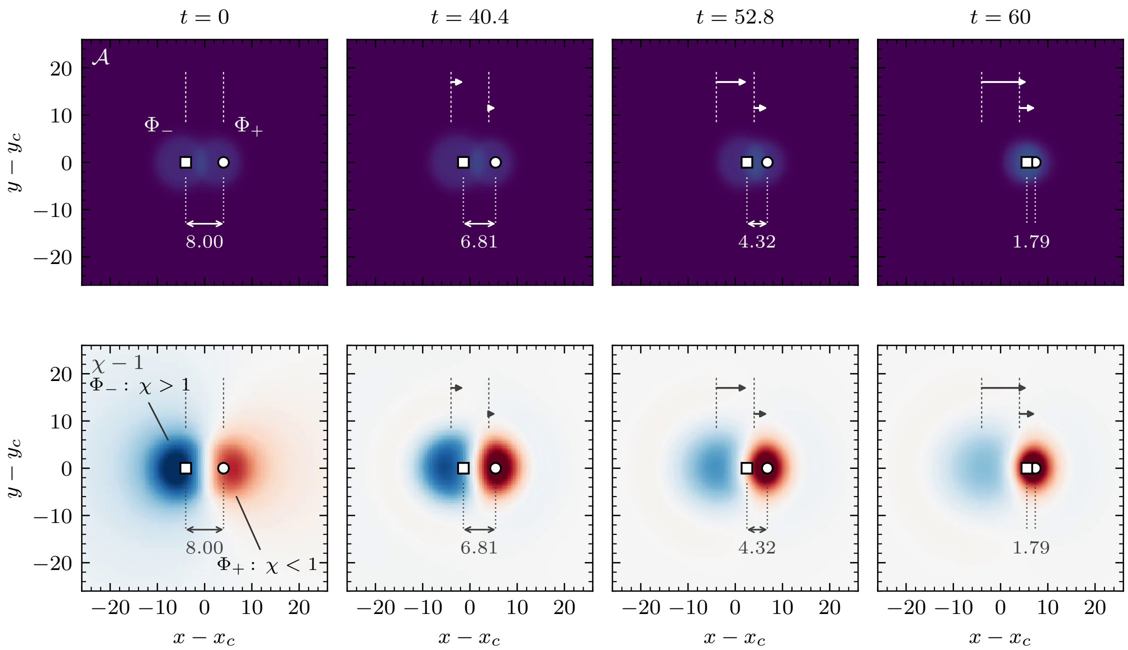

FIG. 3. The runaway, run PM, in the $z = 0$ plane. Top: scalar-field activity $\mathcal{A} = \sum_k (\varphi_k^2 + \Pi_k^2)^{1/2}$, both sectors. Bottom: $\chi - 1$; the phantom star is a hill ($\chi > 1$, blue), the canonical star a well ($\chi < 1$, red). Open markers: measured barycentres (circle canonical, square phantom); dashed verticals mark the release positions $x = \pm 4$, arrows the displacements; brackets give the separation, $8.00 \to 1.79$. Panels are individually color-scaled; the canonical well deepens from $\chi_{\min} = 0.98$ to 0.44.

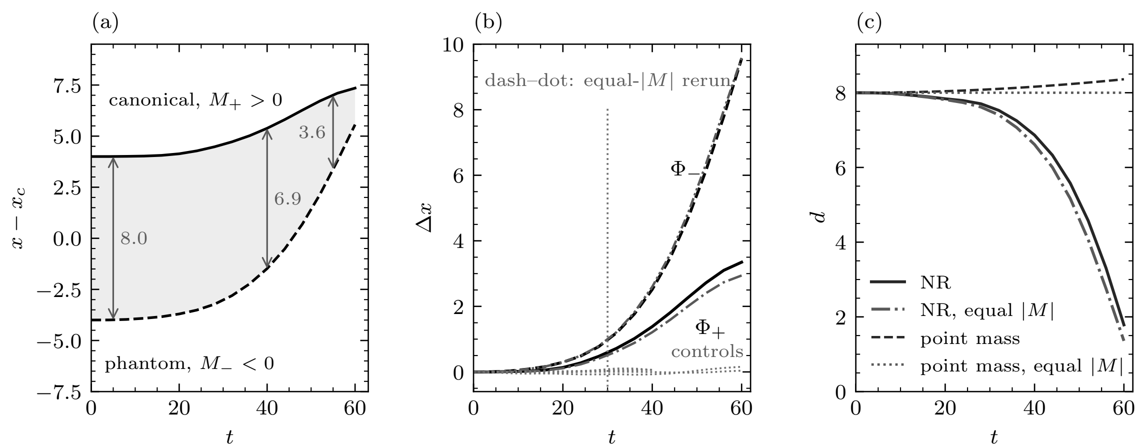

FIG. 4. (a) Barycentre worldlines of run PM: released at rest at $x = \pm 4$, both bodies move in $+x$, the phantom (dashed) chasing the canonical star (solid); the shaded band is the closing gap. (b) Drifts Δx: black solid (Φ_+) and black dashed (Φ_-), reference run; dash–dotted (Φ_+) and dash–dot-dotted (Φ_-), `PM-eq` (Sec. V C); dotted, all four null controls. The dotted line marks the clean-window edge $t = 30$ (Sec. VI). (c) Separation against the matched point-mass integrations: the NR gap closes to near-contact where the model predicts slow growth.

still covers 3.3 times the distance (+2.94 against +9.59; Table I). Theory agrees that the imbalance points the wrong way: each body is accelerated by the *other's* active mass, and $|M_-| > M_+$, so the point-mass model actually gives the *canonical* star the larger early drift (Table II). The asymmetry is an extended-body effect, quantified in Secs. V C and V D: the push on the canonical star stays Newtonian in the phantom's mass to 0.6%, while the phantom's fall runs up to 3× ahead of the point-mass rate because its envelope sits deep inside the canonical star's near field from $t = 0$—a well that deepens further as the canonical star re-accretes its shed field ($\chi_{\min}$: $0.98 \to 0.44$)—whereas the phantom's own shallow, self-diluting hill ($\chi \approx 1.001$) hands the canonical star no matching boost.

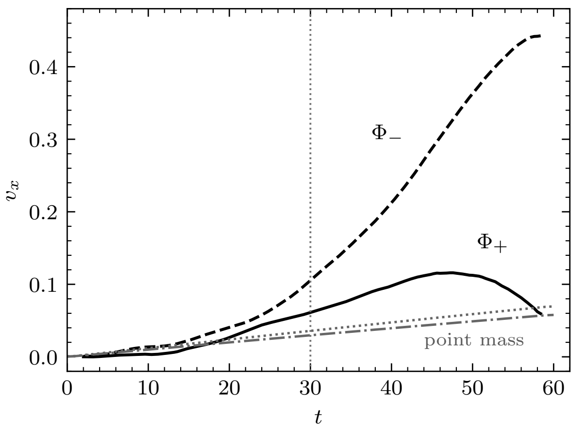

FIG. 5. Barycentre velocities in run PM, in code units ($c = 1$): centred differences of the continuously sampled barycentre stream over ± 2 time units (canonical solid, phantom dashed; the matched point-mass model: canonical dotted, phantom dash–dotted). The dotted line marks the clean-window edge $t = 30$; beyond it the halo bias (Sec. VI) depresses the canonical curve.

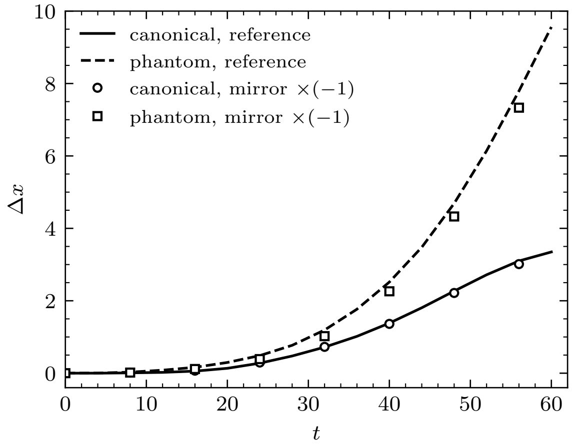

FIG. 6. Mirror control. Lines: reference run PM (canonical solid, phantom dashed); open markers: the mirrored run MP's drift multiplied by -1 (circles canonical, squares phantom). Reflecting the configuration reflects the trajectory to a few percent.

TABLE II. The measured run against the Newtonian idealization. "NR" columns: run PM as measured. "model" columns: a point-mass Bondi integration—two point particles carrying the stars' ADM masses ($+0.0640$ and -0.0770), released at rest at the same separation, each accelerated by the *other's* active mass per Sec. II. Agreement at early times and the growing excess after $t \approx 30$ are discussed in Sec. V D; the clean window is $t \lesssim 30$ (Sec. VI).

	drift Φ_+		drift Φ_-		separation	
t	NR	model	NR	model	NR	model
10	+0.01	+0.06	+0.05	+0.05	7.96	8.01
20	+0.14	+0.24	+0.30	+0.20	7.84	8.04
30	+0.59	+0.54	+0.96	+0.45	7.63	8.09
40	+1.38	+0.96	+2.52	+0.79	6.87	8.16
60	+3.35	+2.13	+9.56	+1.77	1.79	8.36

TABLE III. Does the push scale with its source? Between runs PM-eq and PM exactly one physical input changes: the phantom's mass magnitude, smaller by the factor $0.06395/0.07696 = 0.831$. If the canonical star's motion is gravity sourced by the phantom, its drift must shrink by that same factor. "NR ratio": the measured canonical-star drift of PM-eq divided by that of PM; next to it, the same ratio predicted by the matched point-mass integrations, and the bare mass ratio itself.

t	NR ratio	point-mass ratio	mass ratio
20	**0.836**	0.832	0.831
30	0.859	0.834	0.831
40	0.876	0.837	0.831
60	0.879	0.843	0.831

B. The mirror control closes the artifact objection

A drift produced by grid anisotropy, boundary asymmetry, or diagnostic bias would not know which star is which; swapping the sectors must reverse the physics but not an artifact. It reverses the physics: flipped in sign, run MP reproduces the reference trajectory to 2.1%/15% (canonical/phantom) at $t = 30$ and 3.4%/5.2% at $t = 60$ (Fig. 6).

C. Gravity scales with the source

The strongest quantitative tests compare runs that differ in exactly one quantity.

The push scales with the phantom's active mass. Between PM and PM-eq the phantom's mass changes by the factor 0.831 and nothing else does; the canonical displacement responds in proportion (Table III): drift ratio 0.836 at $t = 20$ (0.6% from the mass ratio 0.831), 0.859 at the window edge $t = 30$ (3.4%)—percent-level agreement across the clean window, degrading slowly beyond it. Barycentre biases largely cancel in this ratio, since both runs share the same separation and numerics.

The fall depends only on the other body. In the same pair of runs the canonical source is unchanged, and the phantom's trajectory with it: drift ratio 1.002 at $t = 20$ and $t = 60$ (1.02–1.03 between). Two independent solves and evolutions, differing in the phantom's own mass by 17%, reproduce its trajectory at the percent level—a physics statement and a reproducibility check at once.

TABLE IV. Ratio of the measured phantom drift to each run's *own* matched point-mass integration. At $d_0 = 8$ the stars overlap from $t = 0$.

d_0	$t = 20$	$t = 30$	$t = 40$
8 (overlapping)	1.48	2.14	3.17
12	1.07	1.36	2.00
16	1.11	1.17	1.48

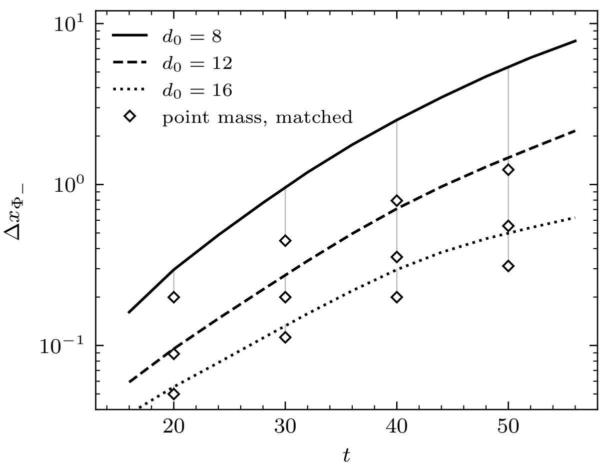

FIG. 7. Separation scaling of the phantom drift (the trustworthy sector; Sec. VI). Curves: NR runs at $d_0 = 8, 12, 16$; diamonds: each run's *own* matched point-mass integration, tied to its curve at equal times (Table IV).

D. Separation scaling: inverse-square once the stars separate

Table II shows the reference run departing from the point-mass model: the phantom overshoots increasingly ($1.5\times$ at $t = 20$, $3.2\times$ at $t = 40$) and the gap closes where the model grows it. `PM-eq` isolates the cause: with $|M|$ matched to 10^{-5} the model predicts identical drifts and a constant gap, yet the NR run gives $+2.94/+9.59$ and a gap closing to 1.35—not a mass-ratio effect. What remains is geometry: at $d_0 = 8$ these stars (rms radius ≈ 5) *overlap from* $t = 0$, the phantom's envelope sitting inside the canonical near field, where the local pull exceeds the point-mass value M/d^2 evaluated at the barycentric separation d.

The wider-separation runs measure this directly (Fig. 7). Fixed-time power-law fits are misleading here—the closing gap makes them report $n = 2.4$–4.6, measuring runaway feedback rather than the force law—so each run is compared against its *own* point-mass integration (Table IV). Where the point-mass idealization applies the measured drift is Newtonian to $\sim 10\%$ at early times; the $d_0 = 8$ excess is the extended-body overlap enhancement, falling away with separation.

TABLE V. Single-star health, $t = 0 \to 40$.

metric	canonical	phantom
extremal χ (min for $+$, max for $-$)	$0.989 \to 0.851$	$1.000 \to 1.001$
rms radius (radiation bath)	$5.05 \to 12.25$	$5.43 \to 6.25$
peak-amplitude breathing	$\pm 8\%$	$\pm 3\%$

E. The phantom sector is the stable one

Counter to intuition, the phantom star was the best-behaved object in the campaign (Table V): gravity is not the binding agent—the identical scalar self-interaction is—and the reversed sign is the *stabilizing* correction. Shed field around the canonical star is pulled back, deepening the well and feeding perturbations (χ $0.99 \to 0.85$ by $t = 40$); around the phantom it anti-gravitates and self-dilutes ($\chi = 1.0009 \pm 0.0002$ all run). The controls agree: `MM` keeps $\chi \approx 1.00$ while its material spreads; `PP` drives χ to 0.44 as it merges.

VI. CAVEATS, AND THEIR MEASURED SCOPE

The barycentre includes the halo. It integrates core plus shed radiation; by $t = 60$ a third of the tracked canonical field has left the accounting region and the distribution has doubled in extent. The clean quantitative window is $t \lesssim 30$, where totals are stable to $\lesssim 7\%$—exactly where the mass-scaling and reproducibility tests are strongest. Late-time drifts are upper bounds on core displacement; the qualitative result holds across the whole run.

The canonical centroid is biased low. At $d_0 = 16$ it drifts *backwards* (-0.34 by $t = 60$): the phantom's repulsion pushes canonical matter out through the $+x$ boundary, and losing the leading edge drags the centroid back. Canonical drifts are therefore underestimates everywhere (including the reference $+3.35$), worst where the true displacement is small—wide gaps, early times. The phantom sector, which gains tracked weight rather than losing it, is the trustworthy one, and Sec. V D uses it; the mass-scaling ratio compares equal-separation runs, so the bias largely cancels there. A core-weighted centroid immune to the halo is built for the follow-up.

The radiation bath cannot leave. Massive-field radiation reflects from massless-wave (Sommerfeld) boundaries and washes back over the stars; this caps the mixed runs at $t = 60$. A sponge layer is the fix.

Resolution. Every run used a single resolution ($N = 128$, $\Delta x = 0.5$: about ten grid points per stellar rms radius), single rank, effectively unigrid (Appendix A). The only sensitivity probe—the $30\times$ tighter constraint solve, which changed nothing—addresses the solve, not the evolution; a proper convergence series is future work.

VII. DISCUSSION AND CONCLUSION

What is established. We posed Bondi's positive–negative mass pair as a full initial-value problem, and it behaved exactly as he argued in 1957. The main results, restated:

(1) *The runaway is real in full general relativity.* Released at rest, the mixed pair—and only the mixed pair—self-accelerates, the phantom chasing the canonical star: displacements +3.35 and +9.56 by $t = 60$, which is 50 to 150 times the single-star noise floor (and at least 20 times every control), and which reverses under mirror reflection to a few percent.

(2) *The effect responds to its source the way gravity must.* The push on the canonical star scales with the phantom's active mass to 0.6–3.4% across the clean window; the phantom's fall is set by the canonical star alone, reproducing across independent solves to 0.2–3%; at wider separations the drift converges to the matched point-mass prediction to about 10%.

(3) *Momentum bookkeeping should balance, and nothing observed contradicts it.* The phantom's momentum enters the signed bookkeeping with a minus sign, so the accelerating pair is expected to carry near-zero total momentum—Forward's accounting carried over to dynamical, radiating, finite-size matter. We state this as a theoretical expectation, guaranteed on shell by the Bianchi identity (Sec. II), not as a measurement: the direct diagnostic—the volume-integrated signed momentum $P_i^{\text{signed}}(t) = \int \big(S_i[\Phi_+] - S_i[\Phi_-]\big)\, d^3x$, decomposed by sector and region—is built but was not run for this campaign, and small momentum-*constraint* norms (Appendix A) are not the same statement.

(4) *A stable star with negative ADM mass exists* as constraint-satisfying, asymptotically flat initial data and survives dynamical evolution—to our knowledge a first in this setting (static phantom-field stars [28] and de Sitter bubbles [26] were known), and possibly the most reusable artifact of the campaign, since the phantom star proved the best-behaved object in it.

What is not established. A *constant-gap* runaway—the textbook picture of the pair accelerating forever at fixed separation—requires point-like separation. At $d_0 = 8$ the stars overlap and close to near-contact by $t = 60$; at $d_0 = 12$–16 the gap-closure is milder but the window ends before the asymptotic behaviour is visible. The clean test needs the wider separations *and* a longer window (a sponge layer), or more compact stars (higher ω, where the sector mass asymmetry also shrinks; Fig. 2). Every mixed run was stopped before contact; what a canonical–phantom collision produces—including whether a horizon forms—is untouched. And nothing here bears on whether NEC-violating matter exists; the model is a probe of the gravitational dynamics, not a proposal for a propulsion system [5].

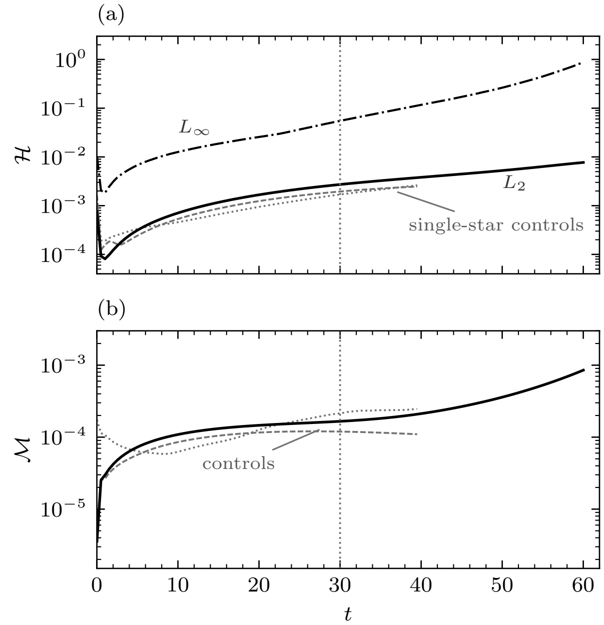

FIG. 8. Constraint monitoring: reference run PM (black) against the single-star controls (gray; dotted canonical, tightly dashed phantom). (a) Hamiltonian constraint $\mathcal{H}$: composite L_2 norms (solid) and the reference run's L_∞ (dash–dotted). (b) Momentum constraint $\mathcal{M}$, L_2. The equal-$|M|$ rerun lies indistinguishably under the black curves and is omitted; the dotted line marks the clean-window edge $t = 30$.

Next steps. Core-weighted centroids and a per-sector momentum-balance diagnostic (the direct Bondi bookkeeping check) are built and waiting; the separation series should be re-run with them before the scaling numbers are quoted as final. A sponge layer unlocks $t \gg 60$ and the constant-gap test; the contact end state and a convergence series follow.

REPRODUCIBILITY

All time series, star tables, parameter files, code patches, analysis scripts, and the campaign journal behind every number and figure are packed, with machine paths scrubbed, in the repository's results tree (`results/bondi-dipole-runaway/`); the figures regenerate from that pack by a single script. Run labels map to pack directories as PM = `pair_pm`, PM-eq = `pair_pm_eqm`, MP = `pair_mp_mirror`, PP = `pair_pp`, MM = `pair_mm`, P = `single_p`, M = `single_m`; the star ODE solver of Sec. III and the $M(\omega)$ scan behind Fig. 2 are `analysis/star_family_scan.py` and the `boson_star_ode` module it drives. Initial data: `GRTresna` [22]; evolution: `GRTeclyn` [21], the AMReX-based [23] GPU port of `GRChombo` [20].

Appendix A: Constraint behaviour

Initial data. `GRTresna`'s nonlinear iteration contracts monotonically on every run; for `PM` the Hamiltonian/momentum error falls 100% → 0.083%/0.074% in seven iterations, and every production run exits at ≤0.09%/≤0.08%—the gate quoted in Table I, in the solver's own grid and normalization.

Evolution. The constraints are re-evaluated on the evolution grid every $\Delta t = 0.5$ and reduced to composite norms over the hierarchy (covered coarse cells dropped, outermost four cells excluded so boundary violation is not misread as interior error). Figure 8 shows the record, in absolute code units: over an overwhelmingly vacuum domain a global *relative* normalization is dominated by near-zero denominators, so the yardstick is the controls run under identical numerics.

Three features carry the message. The birth transient: re-evaluating with fourth-order stencils after the analytic matter reconstruction (Sec. III) gives $L_2(\mathcal{H}) \simeq 7 \times 10^{-4}$, relaxing within $t \lesssim 1$ to the truncation floor 8×10^{-5}. Secular growth: $L_2(\mathcal{H})$ reaches 2.7×10^{-3} at the window edge $t = 30$ and 7.7×10^{-3} at $t = 60$, $L_\infty(\mathcal{H})$ 0.05 and 0.9, concentrated in the deepening canonical well as the pair nears contact and the trapped bath washes back (Sec. VI); $L_2(\mathcal{M})$ stays below 2×10^{-4} through $t = 40$. This growth is the quantitative face of the $t \lesssim 30$ window. And decisively: through that window the mixed run's violation is within a factor 1.6 of the non-drifting single-star controls (2.7×10^{-3} against 1.7×10^{-3} at $t = 30$) while the drifts differ by 50 to 150 times—the drift correlates with the sign structure, not the constraint error.

The single refinement level (curvature-of-χ tagging, $\Delta x \, \|\partial_i \partial_j \chi\| \geq 0.02$, regrid every 16 steps) first activates at $t \approx 47.5$ in the mixed runs and never in the singles. The late trigger is not a schedule but a consequence of the criterion: the stars are weak-field ($|\chi - 1| \lesssim 0.02$ over scales of several m^{-1}), leaving the tagging value an order of magnitude below threshold at $t = 0$, and only the deepening canonical well near contact pushes it across. Every quantitative result was therefore produced on the uniform $\Delta x = 0.5$ grid; forcing refinement on the field profiles from $t = 0$ is part of the planned convergence campaign.

[1] H. Bondi, "Negative mass in general relativity," Rev. Mod. Phys. **29**, 423 (1957).
[2] W. B. Bonnor and N. S. Swaminarayan, "An exact solution for uniformly accelerated particles in general relativity," Z. Phys. **177**, 240 (1964).
[3] J. Bičák, C. Hoenselaers, and B. G. Schmidt, "The solutions of the Einstein equations for uniformly accelerated particles without nodal singularities," Proc. R. Soc. London A **390**, 397 (1983).
[4] W. B. Bonnor, "Negative mass in general relativity," Gen. Relativ. Gravit. **21**, 1143 (1989).
[5] R. L. Forward, "Negative matter propulsion," J. Propulsion Power **6**, 28 (1990).
[6] B. Hoffmann, "Negative mass and the quasars," in *Perspectives in Geometry and Relativity*, edited by B. Hoffmann (Indiana University Press, Bloomington, 1966), p. 176.
[7] R. de A. Martins, "Causal paradoxes implied by the hypothetical coexistence of positive- and negative-mass matter," Lett. Nuovo Cimento **28**, 265 (1980).
[8] R. R. Caldwell, "A phantom menace? Cosmological consequences of a dark energy component with super-negative equation of state," Phys. Lett. B **545**, 23 (2002).
[9] M. S. Morris and K. S. Thorne, "Wormholes in spacetime and their use for interstellar travel: A tool for teaching general relativity," Am. J. Phys. **56**, 395 (1988).
[10] M. Alcubierre, "The warp drive: hyper-fast travel within general relativity," Class. Quantum Grav. **11**, L73 (1994).
[11] R. J. Gleiser and G. Dotti, "Instability of the negative mass Schwarzschild naked singularity," Class. Quantum Grav. **23**, 5063 (2006).
[12] H. Shinkai and S. A. Hayward, "Fate of the first traversible wormhole: Black-hole collapse or inflationary expansion," Phys. Rev. D **66**, 044005 (2002).
[13] R. Friedberg, T. D. Lee, and A. Sirlin, "Class of scalar-field soliton solutions in three space dimensions," Phys. Rev. D **13**, 2739 (1976).
[14] S. Coleman, "Q-balls," Nucl. Phys. B **262**, 263 (1985).
[15] R. Friedberg, T. D. Lee, and Y. Pang, "Mini-soliton stars," Phys. Rev. D **35**, 3658 (1987).
[16] D. J. Kaup, "Klein–Gordon geon," Phys. Rev. **172**, 1331 (1968).
[17] R. Ruffini and S. Bonazzola, "Systems of self-gravitating particles in general relativity and the concept of an equation of state," Phys. Rev. **187**, 1767 (1969).
[18] S. L. Liebling and C. Palenzuela, "Dynamical boson stars," Living Rev. Relativ. **26**, 1 (2023).
[19] D. Alic, C. Bona-Casas, C. Bona, L. Rezzolla, and C. Palenzuela, "Conformal and covariant formulation of the Z4 system with constraint-violating modes," Phys. Rev. D **85**, 064040 (2012).
[20] T. Andrade *et al.*, "GRChombo: an adaptable numerical relativity code for fundamental physics," J. Open Source Softw. **6**, 3703 (2021).
[21] GRTL Collaboration, `GRTeclyn`, https://github.com/GRTLCollaboration/GRTeclyn.
[22] J. C. Aurrekoetxea, S. E. Brady, *et al.*, "GRTresna: an open-source code to solve the initial data constraints in numerical relativity," arXiv:2501.13046 [gr-qc] (2025).
[23] W. Zhang *et al.*, "AMReX: a framework for block-structured adaptive mesh refinement," J. Open Source Softw. **4**, 1370 (2019).
[24] R. Schoen and S.-T. Yau, "On the proof of the positive mass conjecture in general relativity," Commun. Math. Phys. **65**, 45 (1979).

[25] E. Witten, "A new proof of the positive energy theorem," Commun. Math. Phys. **80**, 381 (1981).

[26] S. Mbarek and M. B. Paranjape, "Negative mass bubbles in de Sitter spacetime," Phys. Rev. D **90**, 101502(R) (2014).

[27] J. A. González, F. S. Guzmán, and O. Sarbach, "Instability of wormholes supported by a ghost scalar field. II. Nonlinear evolution," Class. Quantum Grav. **26**, 015011 (2009); see also **26**, 015010 (2009).

[28] V. Dzhunushaliev, V. Folomeev, C. Hoffmann, B. Kleihaus, and J. Kunz, "Boson stars with nontrivial topology," Phys. Rev. D **90**, 124038 (2014).

[29] S. M. Carroll, M. Hoffman, and M. Trodden, "Can the dark energy equation-of-state parameter w be less than -1?," Phys. Rev. D **68**, 023509 (2003).

[30] J. M. Cline, S. Jeon, and G. D. Moore, "The phantom menaced: constraints on low-energy effective ghosts," Phys. Rev. D **70**, 043543 (2004).

[31] T. Helfer, U. Sperhake, R. Croft, M. Radia, B.-X. Ge, and E. A. Lim, "Malaise and remedy of binary boson-star initial data," Class. Quantum Grav. **39**, 074001 (2022).

[32] J. C. Aurrekoetxea, K. Clough, and E. A. Lim, "CTTK: a new method to solve the initial data constraints in numerical relativity," Class. Quantum Grav. **40**, 075003 (2023).

[33] C. Bona, J. Massó, E. Seidel, and J. Stela, "New formalism for numerical relativity," Phys. Rev. Lett. **75**, 600 (1995).

[34] M. Campanelli, C. O. Lousto, P. Marronetti, and Y. Zlochower, "Accurate evolutions of orbiting black-hole binaries without excision," Phys. Rev. Lett. **96**, 111101 (2006).

[35] J. G. Baker, J. Centrella, D.-I. Choi, M. Koppitz, and J. van Meter, "Gravitational-wave extraction from an inspiraling configuration of merging black holes," Phys. Rev. Lett. **96**, 111102 (2006).

[36] N. M. Shirokov, "Wormhole dynamics: nonlinear collapse and gravitational-wave emission," arXiv:2604.00071 [gr-qc] (2026).